

CHAPTER 10

WAVE OPTICS

Syllabus : Wave optics, wavefront and Huygen's wave theory, using wavefront reflection and refraction of light on plane surface, verification of laws of reflection and refraction using Huygen's wave theory, interference, Young's double slit experiment and expression for fringe width, coherent sources and sustained interference, diffraction of light due to single slit and width of central maxima. Polarization, plane polarized light, plane polarized light and uses of polaroids.

POINTS TO REMEMBER

- ★ As per principle of superposition of waves, the resultant displacement at a point in a medium is given by :

$$\vec{y} = \vec{y}_1 + \vec{y}_2 + \dots + \vec{y}_n.$$

- ★ Conditions for constructive interference of light :

Phase difference $\phi = 2n\pi$ i.e., even multiple of π .

Path difference $\Delta = 2n\frac{\lambda}{2}$ i.e., even multiple of $\frac{\lambda}{2}$.

- ★ Conditions for destructive interference of light.

Phase difference $\phi = (2n-1)\pi$ i.e., odd multiple of π .

Path difference $\Delta = (2n-1)\frac{\lambda}{2}$ i.e., odd multiple of $\frac{\lambda}{2}$.

- ★ In interference of light : Resultant amplitude $R = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \phi}$

$$\frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2}.$$

- ★ In Young's double slit experiment, the fringe width $\beta = \frac{\lambda D}{d}$.
- ★ The bending of light round the corners of the obstacle so that light encroaches the geometrical shadow is called diffraction.
- ★ The lack of symmetry of the light waves along the direction of propagation of light is called polarisation.

Objective Type Questions

Q. 1. Choose the Correct Answer :

1. A light ray appears to travel in a straight path because :

- (a) It is made up of tiny particles (b) Its velocity is high
(c) Its wavelength is very small (d) None of the above.

2. Which of the following phenomenon does not give evidence of wave nature of light :
 (a) Refraction (b) Reflection (c) Interference (d) Polarisation.
3. Two coherent sources of light must have the same :
 (a) Amplitude (b) Phase difference (c) Frequency (d) Both (b) and (c).
4. Which of the following wave can not be polarised :
 (a) X-rays (b) Radio waves
 (c) Longitudinal waves (d) Transverse waves.
5. The intensity of central fringe in Young's experiment is I . One slit is closed then intensity becomes I_0 . Which of the following is true :
 (a) $I = I_0$ (b) $I = 2I_0$
 (c) $I = 4I_0$ (d) No relation between I and I_0 .
6. A colourless soap solution in white light appears to be coloured because of :
 (a) Reflection (b) Refraction (c) Interference (d) Diffraction.
7. The ratio of amplitudes of light waves is 2 : 3. Due to their interference, ratio of maximum and minimum intensity of light will be :
 (a) 2 : 3 (b) 4 : 9 (c) 5 : 1 (d) 25 : 1.
8. Young's double-slit experiment proves that :
 (a) Light rays consist of particles
 (b) Light rays are waves
 (c) Light rays are neither particles nor waves
 (d) None of these.
9. Which of the following cannot be explained by wave nature of light :
 (MP 2009 Set B)
 (a) Polarisation (b) Interference
 (c) Diffraction (d) Photoelectric effect.
10. Waves are carrier of :
 (a) Energy (b) Amplitude (c) Wavelength (d) Matter.

Ans. 1. (c), 2. (b), 3. (c), 4. (c), 5. (c), 6. (d), 7. (d), 8. (b), 9. (d), 10. (a).

Q. 2. Fill in the Blanks :

1. The light is transverse wave, it is proved by of light.
2. In Young's double slit experiment, the distance between two successive bright fringes is known as
3. The phase difference between two coherent sources of light is
4. In constructive interference of light path difference is always multiple of $\frac{\lambda}{2}$.
5. Young's double-slit experiment proves theory of light.
6. Bending of waves around the corners of an obstacle is known as
7. The phase difference between two points situated in the same wave front is
8. At polarising angle, the angle between reflected and refracted ray is
 (MP 2012)
9. The surface of medium at an instant of time, in which all the particles of medium vibrates with same phase is called

- (vi) Phase difference of the two waves should be zero or should be constant.
- (vii) Light sources should be monochromatic.
- (viii) If light rays are polarized, then their plane of polarization should be parallel.
- (ix) Both the sources should be coherent (*i.e.*, phase difference between the two waves should be zero or constant).

***Q. 2. What is the difference between interference and diffraction ?**

(MP 2000, 03, 08 Set C, 09 Set D, 13, 14, 16)

Ans. Difference between interference and diffraction :

Interference	Diffraction
1. Interference fringes are formed when the two light rays from the coherent sources are superimposed.	1. Fringes are formed due to superimposition of light rays from the same source.
2. All bright fringes are equally illuminated.	2. Central fringe is brightest and all other fringes have intensity in decreasing order.
3. Width of all fringes are always equal.	3. Fringe width decreases when we move away from the centre.
4. Minimum intensity is zero, so fringes can be distinguished very well.	4. Minimum intensity fringes are not perfectly darker so it is relatively difficult to distinguish bright and dark fringes.

***Q. 3. Explain Huygen's principle of secondary wavelets.**

(MP 2010, 12)

Ans. Huygen's secondary wavelets theory : The main postulates of this theory are given ahead :

(i) When a disturbance is created in a medium the particles of the medium starts vibrating. The surface in the medium, on which each particle vibrate in same phase is called the wave front. If the source is like a point, at a finite distance, the wave front is always spherical. If the source is in form of a slit at a finite distance the wave front is cylindrical.

(ii) Every point of the medium on a given wave front is a source of originating new disturbances. These new disturbances are called secondary wavelets.

(iii) These secondary wavelets advances in the medium with same velocity as of original wave.

(iv) The common tangential surface drawn on these secondary wavelets at any instant, shows the position of new wave front at that instant.

Q. 4. What is polaroid ? State its applications in daily life.

(MP 2000, 06)

Or

What is polaroid ? Explain polaroid on the basis of the following points :

(a) Working, (b) Uses (any two).

(MP 2011)

Ans. Polaroid : Polaroid are large sized manufactured polarizing films capable of producing plane polarized beam of light.

(a) **Working :** The polaroid has a characteristic plane called transmission plane. When an unpolarized light falls on it only the vibrations of electric vector, parallel to the plane of transmission get transmitted. Thus the transmitted light obtained is plane polarized.

(b) **Uses :** (i) These are used in laboratory to produce and analyse plane polarized light.

(ii) These are used in sun glasses.

(iii) These are used in headlights and wind screen of automobile to cut-off the dazzling light.

(iv) These are used for viewing three dimensional pictures.

Q. 5. What do you understand by plane of vibration and plane of polarization in case of plane polarized light ? Draw diagram to explain your answer.

(MP 2000, 02, 09 Set B)

Or

Define : (a) Plane of vibration, (b) Plane of polarisation.

(MP 2017)

Ans. Plane of polarization : The plane which does not contain the component of vibration of polarized beam is called plane of polarization. Plane EFGH is plane of polarization.

Plane of vibration : The plane containing the incident ray and parallel to crystallographic axis is called plane of vibration. Plane of polarization is always perpendicular to plane of vibration.

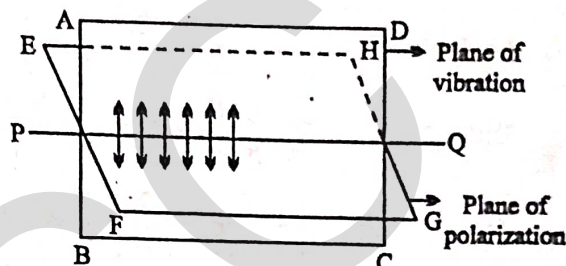


Fig.

Q. 6. Answer the following questions :

(NCERT)

(a) In a single slit diffraction experiment, the width of the slit is made double the original width. How does this affect the size and intensity of the central diffraction band ?

(b) In what way is diffraction from each slit related to interference pattern in a double slit experiment ?

(c) When a tiny circular obstacle is placed in the path of light coming from a distant source a bright spot is seen at the centre of the shadow of the obstacle. Explain why ?

(d) Two students are separated by a 7 m partition wall in a room 10 m high. If both light and sound waves can bend round corners, how is it that the students are unable to see each other even though they can converse easily.

(e) Ray optics is based on the assumption that light travels in a straight line. Diffraction effects (observed when light propagates through small apertures/slits or around small obstacles) disprove this assumption. Yet the ray optics assumption is so commonly used in understanding location and several other properties of images in optical instruments. What is the justification ?

Sol. (a) Width of central diffraction band $\beta_0 = \frac{2\lambda}{a}$ i.e., $\beta_0 \propto \frac{1}{a}$, where a is slit width.

Therefore, on doubling the slit width, the width of central bright band will be halved but the intensity will increase four times.

(b) If width of each slit is of the order of λ , then interference pattern in double slit experiment is modified by the diffraction pattern from each of the two slits.

(c) When a small circular obstacle is placed in the path of light coming from a distant source, then the light waves diffract from the edges of obstacle. Diffract rays produce constructive interference at the centre of the shadow and hence bright spot is formed.

(d) For diffraction of waves by obstacle/aperture, through a large angle, the size of obstacle/aperture should be comparable to wavelength.

Wavelength of sound is of the order of size of obstacle (partition wall). Therefore, sound waves diffract easily and reach to the other student and they converse easily wavelength of light waves is too small than the size of obstacle. Therefore, light waves do not diffract. Thus, students cannot see each other.

(c) The size of aperture of lenses used in optical instruments is too large as compared to the wavelength of light used. Therefore, diffraction is negligible. That is why the position of images and its characteristics are studied on the concept of ray optics.

Numerical Questions

Q. 1. In a Young's double slit experiment of the fringe width is 4 mm. Obtain distance of third dark fringe from central bright fringe.

Sol. Given : $\beta = 4\text{mm} = 4 \times 10^{-3}\text{m}$

For third dark fringe $n = 3$

Formula : $x_n = \frac{(2n-1)}{2} \cdot \frac{\lambda D}{d}$

But $\frac{\lambda D}{d} = \beta$

$$\therefore x_n = \frac{(2n-1)}{2} \beta = \left[\frac{2 \times 3 - 1}{2} \right] \times 4 \times 10^{-3}\text{m}$$

$$\Rightarrow x_n = 10 \times 10^{-3}\text{m} = 10^{-2}\text{m} = 10\text{mm.}$$

Ans.

Q. 2. Monochromatic light of wavelength 589 nm is incident on air to water interface. What will be the wavelength, frequency and speed of (a) reflected light (b) refracted light ? Refractive index of water is 1.33. (NCERT)

Sol. Given, $\lambda = 589\text{nm} = 589 \times 10^{-9}\text{m}$, $\mu = 1.33$

(a) For reflected light : In reflection, there is no change of medium. Therefore, wavelength, frequency and speed will not change.

Thus, $\lambda = 589\text{nm}$; Speed $c = 3 \times 10^8\text{m/s}$

$$\text{and frequency } \nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{589 \times 10^{-9}} = 5.09 \times 10^{14}\text{Hz.}$$

Ans.

(b) For refracted light :

Frequency remains the same but speed and wavelength both will change

$$\text{Speed } v = \frac{c}{\mu} = \frac{3 \times 10^8}{1.33} = 2.26 \times 10^8\text{m/s}$$

$$\text{Wavelength } \lambda' = \frac{v}{\nu} = \frac{2.26 \times 10^8}{5.09 \times 10^{14}} = 0.444007 \times 10^{-6}$$

$$= 444 \times 10^{-9}\text{m} = 444\text{nm.}$$

Ans.

Q. 3. In Young's double slit experiment, the slits are separated by 0.25 mm and the screen is placed 1.4 m away. The distance between the central bright fringe and the fourth bright fringe is measured to be 1.2 cm. Determine the wavelength of the light used. (NCERT)

Sol. Given, $d = 0.25\text{mm} = 0.25 \times 10^{-3}\text{m}$; $D = 1.4\text{m}$;

$$x = 1.2\text{cm} = 1.2 \times 10^{-2}\text{m}; n = 4$$

For constructive interference $x = \frac{n\lambda D}{d}$

$$\lambda = \frac{x \cdot d}{n \cdot D}$$

$\therefore$

$$\Rightarrow \lambda = \frac{1.2 \times 10^{-2} \times 0.25 \times 10^{-3}}{4 \times 1.4} = 5357 \times 10^{-7} \text{ m}$$

$$\Rightarrow \lambda = 536 \text{ nm.}$$

Ans.

Q. 4. A beam of light consisting two wavelengths 650 nm and 520 nm, is used to obtain interference fringes in Young's double slit experiment.

(a) Find the distance of the third bright fringe on the screen from the central maxima for wavelength 650 nm.

(b) What is the least distance from the central maxima, where the bright fringes due to both the wavelengths coincide? $d = 2\text{ mm}$; $D = 120 \text{ cm}$. (NCERT)

Sol. Given, $\lambda_1 = 650 \text{ nm} = 650 \times 10^{-9} \text{ m}$

$$\lambda_2 = 520 \text{ nm} = 520 \times 10^{-9} \text{ m}$$

$$(a) \quad x = \frac{n\lambda_1 D}{d} = \frac{3 \times 650 \times 10^{-9} \times 120 \times 10^{-2}}{2 \times 10^{-3}} \\ = 117000 \times 10^{-8} \text{ m} = 1.17 \times 10^{-3} \text{ m} = 1.17 \text{ mm.}$$

Ans.

(b) Let at distance x from the central maxima bright fringes of both the wavelengths coincide.

$$\text{Then, } x = \frac{n_1 \lambda_1 D}{d} = \frac{n_2 \lambda_2 D}{d} \text{ or } n_1 \lambda_1 = n_2 \lambda_2 \quad \dots(1)$$

$$\frac{\lambda_1}{\lambda_2} = \frac{n_2}{n_1}$$

$$\lambda_1 > \lambda_2 \Rightarrow n_2 > n_1$$

$$\therefore n_2 = (n_1 + 1)$$

Putting value in eqn. (1),

$$n_1 \lambda_1 = (n_1 + 1) \lambda_2$$

$$n_1 \times 650 = (n_1 + 1) 520$$

$$5n_1 = 4n_1 + 4 \Rightarrow n_1 = 4$$

$$\therefore x = \frac{n_1 \lambda_1 D}{d} = \frac{4 \times 650 \times 10^{-9} \times 120 \times 10^{-2}}{2 \times 10^{-3}}$$

$$\Rightarrow x = 156000 \times 10^{-8} \text{ m} = 1.56 \times 10^{-3}$$

$$\Rightarrow x = 1.56 \text{ mm.}$$

Ans.

Q. 5. In double slit experiment using light of wavelength 600 nm, the angular width of a fringe formed on a distant screen is 0.1° . What is the spacing between the two slits? (NCERT)

Sol. Given, $\lambda = 600 \text{ nm} = 600 \times 10^{-9} \text{ m} = 6 \times 10^{-7} \text{ m}$

$$\theta = 0.1^\circ = \frac{0.1\pi}{180} \text{ rad.}$$

Using $\theta = \frac{\lambda}{d},$

$$d = \frac{\lambda}{\theta} = \frac{6 \times 10^{-7}}{\frac{\pi}{180} \times 0.1} = 3.44 \times 10^{-4} \text{ m.}$$

Ans.

Q. 6. A parallel beam of light of wavelength 500 nm falls on a narrow slit and the resulting diffraction pattern is observed on a screen 1 m away. It is observed that the first minimum is at a distance of 2.5 mm from the centre of the screen. Find the width of the slit. (NCERT)

Sol. Given, $\lambda = 500 \text{ nm} = 500 \times 10^{-9} \text{ m}; = 5 \times 10^{-7} \text{ m}$

$$D = 1 \text{ m}; x = 2.5 \text{ mm} = 2.5 \times 10^{-3} \text{ m}$$

The condition for minima is $a \frac{x}{D} = n\lambda$

$$\therefore a = \frac{n\lambda D}{x} = \frac{1 \times 5 \times 10^{-7} \times 1}{2.5 \times 10^{-3}}$$

$$\Rightarrow a = 2 \times 10^{-4} \text{ m} = 0.2 \text{ mm.}$$

Ans.

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